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Mid-Defense Report

ON

PFM(Personal Finance Manager)

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ABSTRACT

The Personal Finance Manager (PFM) is a web-based application developed to simplify personal expense tracking through automation and intelligent financial analysis. Traditional budgeting applications often fail because users must manually enter transaction details using repetitive forms, leading to frustration and user abandonment over time. PFM addresses this problem by allowing users to record expenses using simple natural language text such as “paid 500 for lunch,” which is automatically processed using a custom Natural Language Processing (NLP) parser based on regular expressions and probabilistic keyword matching. The system extracts transaction amounts, determines spending categories, and stores the information without requiring manual form-filling. When the local parser cannot confidently classify a transaction, the system utilizes the NIM API for AI-assisted categorization. In addition to expense tracking, the application includes a greedy budget optimization algorithm that analyzes historical spending patterns and recommends category-wise spending reductions to help users achieve savings goals. The system also implements a Retrieval-Augmented Generation (RAG) module that enables users to ask natural language questions about their financial history and receive grounded responses based strictly on their stored transaction records. The application is developed using React for the frontend, FastAPI for backend services, and PostgreSQL through Supabase for database management and authentication. Overall, the project demonstrates the practical integration of NLP, AI-assisted financial analysis, and optimization algorithms within a modern full-stack web application to reduce the friction associated with personal finance management.

Keywords: Personal Finance, Expense Tracking, Natural Language Processing, Budget Optimization, RAG, FastAPI, React, Supabase.

Table of Contents

ACKNOWLEDGEMENT.....	ii
ABSTRACT	iii
Table of Contents	iv
LIST OF FIGURES	vi
LIST OF TABLES	vii
LIST OF ABBREVIATION.....	viii
CHAPTER 1: INTRODUCTION.....	1
1.1 Introduction	1
1.2 Problem of Statement	1
1.3 Project Objectives	2
1.4 Scope and Limitation	2
1.5 Report Organization	2
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW	4
2.1 Background Study.....	4
2.2 Terminologies Used.....	4
2.3 Literature Review	5
CHAPTER 3: SYSTEM ANALYSIS AND DESIGN	6
3.1 System Analysis	6
3.1.1 Analysis Model	6
3.1.2 Requirement Analysis.....	7
3.1.3 Feasibility Analysis	10
3.1.4 Object Modelling: Object & Class Diagram	11
3.1.5 Dynamic Modelling: State & Sequence diagram	13
3.2 System Design.....	15
3.2.1 Component Diagram.....	15
3.3 Algorithm Details	15
CHAPTER 4: IMPLEMENTATION AND TESTING	18

4.1 Implementation	18
4.1.1 Tools Used.....	18
4.1.2 Implementation Details of Modules.....	19
4.2 Testing.....	19
4.2.1 Test Cases for Unit Testing	20
CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION.....	22
5.1 Modules Completed	22
5.2 Modules Remaining	22
References.....	24

LIST OF FIGURES

Figure 3. 1: Agile Methodology of "PFM"	7
Figure 3. 2: Use Case Diagram of "PFM"	9
Figure 3. 3: Gantt Chart Diagram of "PFM"	11
Figure 3. 4: Class Diagram of "PFM"	12
Figure 3. 5: Object Diagram of "PFM"	12
Figure 3. 6: State Diagram of "PFM"	13
Figure 3. 7: Sequence Diagram of "PFM"	13
Figure 3. 8: Activity Diagram of "PFM"	14
Figure 3. 9: Componenet Diagram of “PFM”	15

LIST OF TABLES

Table 1: Unit Test of ‘PFM’	21
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LIST OF ABBREVIATION

AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
NLP	Natural Language Processing
PFM	Personal Finance Manager
PostgreSQL	Postgres Structured Query Language
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
SDK	Software Development Kit
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language

URL Uniform Resource Locator

VS Code Visual Studio Code

CHAPTER 1: INTRODUCTION

1.1 Introduction

Personal finance management is one of the most important aspects of modern life. Individuals need to monitor their income, expenses, and savings in order to maintain financial stability and achieve long-term goals. Although many budgeting and expense tracking applications exist, most users abandon these systems after a short period because of the repetitive and time-consuming process of manual data entry.

Most traditional applications require users to manually fill out forms, select categories, choose dates, and save each transaction individually. While this process appears simple initially, it becomes frustrating over time, especially for users with frequent daily expenses. As a result, users often stop tracking their finances completely.

The Personal Finance Manager (PFM) project was developed to solve this usability problem. Instead of relying on traditional form-based inputs, the application allows users to log expenses using simple natural language text. For example, a user can type “paid 500 for lunch” or “spent 2000 on shopping,” and the system automatically extracts the amount and categorizes the expense.

The project combines Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and budget optimization algorithms to provide users with a smarter and more interactive financial management experience. Alongside expense tracking, the system also analyzes spending habits and recommends budget reductions based on historical data.

The application is designed as a modern full-stack web application using React, FastAPI, and Supabase. The system focuses on simplicity, automation, and user convenience while maintaining secure data management.

1.2 Problem of Statement

Despite the availability of many personal finance applications, users often fail to maintain consistent financial tracking because most systems require repetitive manual data entry and complex form handling for every transaction. Existing applications also struggle to understand the context of user expenses, which can lead to incorrect categorization and

inaccurate financial records. Additionally, most budgeting tools lack intelligent financial analysis and personalized budget recommendations based on actual spending behavior. Therefore, there is a need for a smart financial management system that simplifies expense tracking through natural language input, provides automatic categorization, offers intelligent budget optimization, and enables conversational financial analysis.

1.3 Project Objectives

The objective of the project named as PFM is to develop a smart Personal Finance Manager that automates expense tracking and budget analysis using natural language processing.

1.4 Scope and Limitation

Scope: The scope of this project is to develop a web-based Personal Finance Manager that allows users to record expenses using natural language input, automatically categorize transactions, analyze spending patterns, and provide intelligent budget optimization suggestions. The system also includes conversational financial analysis through AI-based retrieval techniques and offers a simple, user-friendly interface for efficient personal finance management.

Limitations: The project has several limitations, including the absence of direct bank or payment gateway integration, dependence on manually entered expense data, and possible inaccuracies when processing unclear or ambiguous text inputs. Additionally, advanced features such as multi-currency support, tax calculation, receipt scanning, and official financial report generation are not included within the scope of this project.

1.5 Report Organization

In order to provide a clear picture of the writing of this project, a report organization has been compiled containing information on the material discussed in each chapter such that:

- **Chapter 1: Introduction**

This chapter contains all the introduction part which describes the idea of the project. It includes the problem statement, objectives, scope and limitation.

- **Chapter 2: Background Study and Literature Review**

This chapter contains the background study about the system and literature review part in which study of existing similar of project is done.

- **Chapter 3: System Analysis and Design**

This chapter includes system analysis of the project where different requirement of the project is analyzed.

- **Chapter 4: Implementation and Testing**

This chapter includes the details of implementing, testing phases of the system.

- **Chapter 5: Conclusion and Future Recommendation**

This chapter includes the summarization of project, what type of outcome is generated from project and the future changes.

Lastly, references and the log sheet provided by the supervisor during the week visit.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Personal finance management systems are designed to help users track expenses, manage budgets, and improve financial planning. Although many financial tracking applications are available today, most of them rely heavily on manual data entry, requiring users to repeatedly fill forms and select categories for every transaction. This repetitive process often reduces user engagement and leads people to stop tracking their finances regularly. To address this issue, modern systems are increasingly integrating intelligent technologies that simplify expense management and improve user convenience.

The Personal Finance Manager (PFM) project uses technologies such as Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and budget optimization algorithms to create a smarter financial management system. NLP enables the system to understand user-entered expense descriptions and automatically extract important transaction details, while RAG allows users to ask conversational questions based on their stored financial records. The system also analyzes spending patterns and provides budget optimization suggestions to help users manage their finances more effectively. The application is developed using React, FastAPI, and Supabase to ensure a modern, secure, and responsive web-based platform.

2.2 Terminologies Used

This project involves several important terminologies related to web development, artificial intelligence, and financial management systems. Personal Finance Management refers to the process of tracking income, expenses, savings, and budgeting activities. Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand and process human language inputs. Retrieval-Augmented Generation (RAG) is an AI technique that retrieves relevant information from a database before generating responses, ensuring accurate and context-based answers. FastAPI is a Python framework used for developing backend APIs, while React is a JavaScript library used for building interactive user interfaces. Supabase is an open-source backend platform that provides PostgreSQL database services and authentication. Budget Optimization refers to the

process of analyzing spending patterns and suggesting financial adjustments to help users achieve savings goals efficiently.

2.3 Literature Review

Various studies and existing applications related to personal finance management were reviewed during the development of the Personal Finance Manager (PFM) system. Existing applications such as Splitwise focus mainly on bill splitting and shared expense management among groups. Although the application simplifies shared payments, it lacks features such as personal budget optimization, intelligent financial analysis, and automatic expense categorization through natural language input[1].

Another widely used budgeting system is YNAB, which encourages users to assign every dollar to a specific category for better financial planning. The system provides detailed budgeting tools and financial reports; however, it depends heavily on manual transaction entry and continuous user involvement. This creates usability challenges for users who prefer quicker and simpler expense tracking methods[2].

Wallet is another popular personal finance management application that provides expense tracking, budgeting, and spending analytics. While the application offers useful visual reports and synchronization features, users still need to manually manage transaction records and categories. The system does not provide conversational expense entry or AI-assisted financial interaction[3].

Research conducted by Fabrizio Sebastiani on automated text categorization demonstrated how Natural Language Processing and machine learning techniques can classify textual inputs effectively using token analysis and probabilistic methods. This study provided theoretical support for implementing the NLP-based expense categorization mechanism in the proposed system[4].

Similarly, studies on Retrieval-Augmented Generation (RAG) highlighted the importance of grounding AI-generated responses using external databases to improve accuracy and reduce hallucination. These concepts influenced the development of the conversational financial assistant in PFM, where user financial queries are answered using actual transaction records stored in the database rather than generalized AI responses[5].

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

System analysis is the process of studying and understanding the requirements, functionality, and overall behavior of a proposed system before implementation. The main purpose of system analysis is to identify user needs, analyze existing problems, and define the functional and non-functional requirements of the system. In the Personal Finance Manager (PFM) project, system analysis was performed to understand the limitations of existing personal finance applications and design a system that simplifies expense tracking through automation and intelligent financial analysis.

The analysis identified that most existing financial management systems rely heavily on manual data entry, repetitive form handling, and fixed expense categorization methods, which often reduce user engagement and lead to inaccurate financial tracking. To overcome these issues, the proposed system focuses on natural language expense entry, automatic expense categorization, conversational financial analysis, and intelligent budget optimization. The system is designed as a web-based platform where users can easily manage their financial activities through a simple and interactive interface.

The analysis phase also helped identify the technologies, modules, and workflows required for the successful implementation of the system. The major modules identified during system analysis include user authentication, expense management, NLP-based expense parsing, budget optimization, dashboard analytics, and Retrieval-Augmented Generation (RAG)-based financial assistance. The findings from the system analysis phase served as the foundation for the system design and implementation process.

3.1.1 Analysis Model

The analysis model for the "PFM(Personal Finance Manager)" follows the Agile methodology, which focuses on iterative development and flexibility. Agile ensures that the system can be adapted efficiently, making it easier to implement changes and meet expectations throughout the development process.

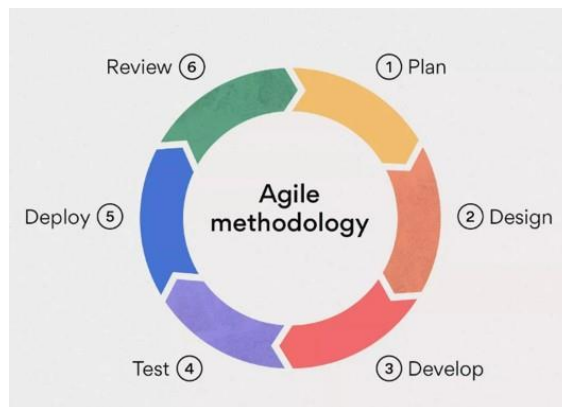


Figure 3. 1: Agile Methodology of "PFM"

Key Phases in Agile Methodology:

1. **Planning:** Defines the project scope, objectives, and deliverables. Identifies key features such as user registration, doctor directory, appointment scheduling.
2. **Design:** Create wireframes to visualize the user interface and user experience.
3. **Development:** Implements PFM features like expense tracking, NLP parsing, budget optimization, and RAG using React, FastAPI, Supabase, and NIM API.
4. **Testing:** Conducts testing after each iteration to identify and fix bugs, ensuring the system's reliability.
5. **Deployment:** Releases the system to a staging environment for user acceptance testing. After successful validation, deploys the system to the production environment.
6. **Review & Maintenance:** Continuously monitor the system's performance, gathers user feedback, and implementing necessary updates and improvements.

3.1.2 Requirement Analysis

Requirement analysis is the initial tasks to be performed while developing the system.

3.1.2.1 Functional Requirement

Functional requirements for the Personal Finance Manager (PFM) define the system behavior to support users in managing their personal finances efficiently.

- **Registration and Login:** Users can register, log in, and manage their profiles securely for personalized access to the system.
- **Expense Management:** Users can add, edit, delete, and view expenses using natural language input for easier financial tracking.
- **Automatic Expense Categorization:** The system automatically extracts transaction details and categorizes expenses using NLP techniques.
- **Budget Optimization:** Users receive intelligent budget suggestions based on their spending history and savings goals.
- **Financial Analysis and Dashboard:** Users can view spending summaries, category-wise analytics, and financial trends through interactive dashboards.
- **Conversational Financial Query:** Users can ask natural language questions related to their expenses and receive context-aware responses through the RAG module.

3.1.2.2 Use Case Diagram

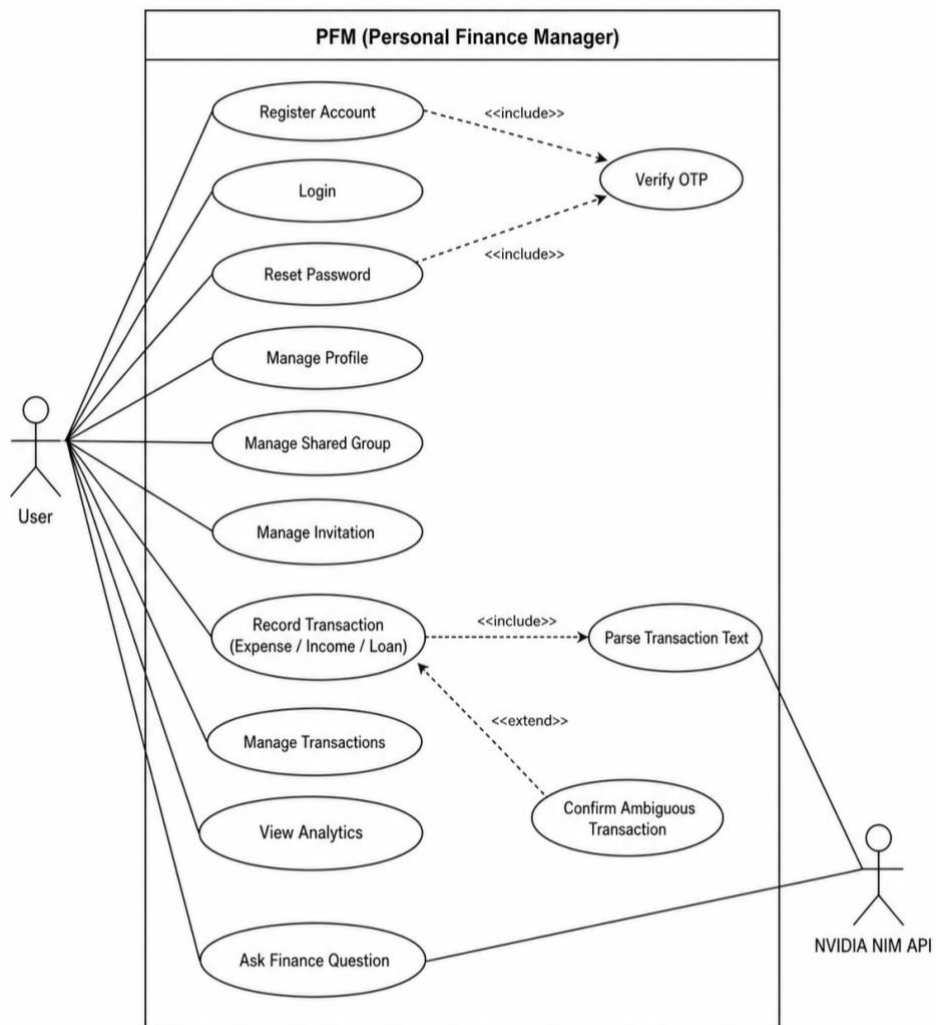


Figure 3. 2: Use Case Diagram of "PFM"

3.1.2.3 Non Functional Requirement

Non-functional requirements for the Personal Finance Manager (PFM) define the quality attributes of the system, focusing on usability, performance, security, and reliability for efficient financial management.

- **Usability:** The system provides a simple, intuitive, and user-friendly interface that allows users to easily manage expenses, view financial analytics, and interact with the application using natural language input.

- **Responsiveness:** The system is designed as a responsive web application using React, ensuring smooth accessibility across desktops, tablets, and mobile devices.
- **Performance:** The system processes expense entries, NLP categorization, and financial queries efficiently with minimal response time while securely storing and retrieving data from Supabase PostgreSQL databases.
- **Security:** The system ensures secure user authentication, protected financial data, and isolated user records through Supabase authentication and database security mechanisms.
- **Availability:** The application is designed to remain accessible at all times, allowing users to manage expenses, analyze budgets, and access financial information without interruption.

3.1.3 Feasibility Analysis

The feasibility study is conducted to determine whether the proposed Personal Finance Manager (PFM) project is practical and achievable based on technical, economic, operational, and schedule-related factors. This analysis helps evaluate whether the system can be successfully developed and implemented using the available resources and technologies.

Technical Feasibility: The project is technically feasible because the required technologies such as React, FastAPI, Supabase, PostgreSQL, and NIM API are available and suitable for developing the system. The existing hardware and software resources are sufficient to support the development and deployment of the application.

Economic Feasibility: The project is economically feasible as it mainly uses open-source technologies and free-tier cloud services, reducing development and deployment costs. No significant investment in additional hardware or manpower is required.

Operational Feasibility: The system is operationally feasible because it provides a simple and user-friendly interface that allows users to manage expenses easily using natural language input. Although technical knowledge is required for the development team, users do not need advanced technical skills to operate the system.

Schedule Feasibility: The project is schedule feasible because the development tasks can be completed within the allocated academic timeline using Agile methodology. Proper

planning and resource management help ensure the timely completion of system development, testing, and deployment.



Figure 3. 3: Gantt Chart Diagram of "PFM"

3.1.4 Object Modelling: Object & Class Diagram

Object Modelling in the Personal Finance Manager (PFM) represents the structure of the system using classes and their relationships. The system includes major entities such as User, Expense, Category, Budget, Transaction, and NLPParser. A User can manage multiple expenses, while the Expense class stores details like amount, category, description, and date. The Category class organizes expenses into groups such as food, shopping, and utilities, and the Budget class manages spending analysis and savings goals. The NLPParser processes natural language expense inputs and automatically extracts transaction details. These classes work together to support features like user authentication, expense tracking, automatic categorization, budget optimization, and conversational financial analysis, creating a structured and scalable system architecture.

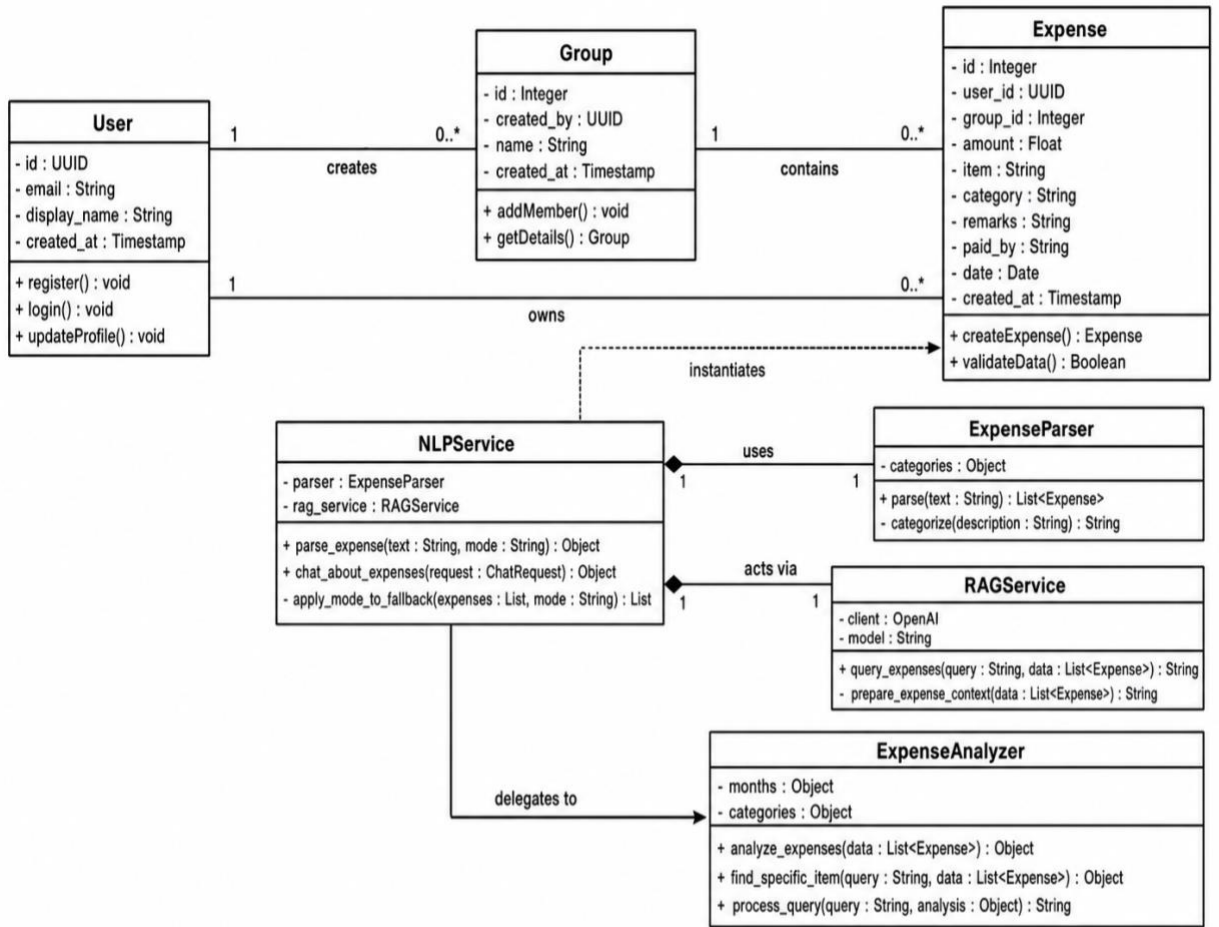


Figure 3. 4: Class Diagram of "PFM"

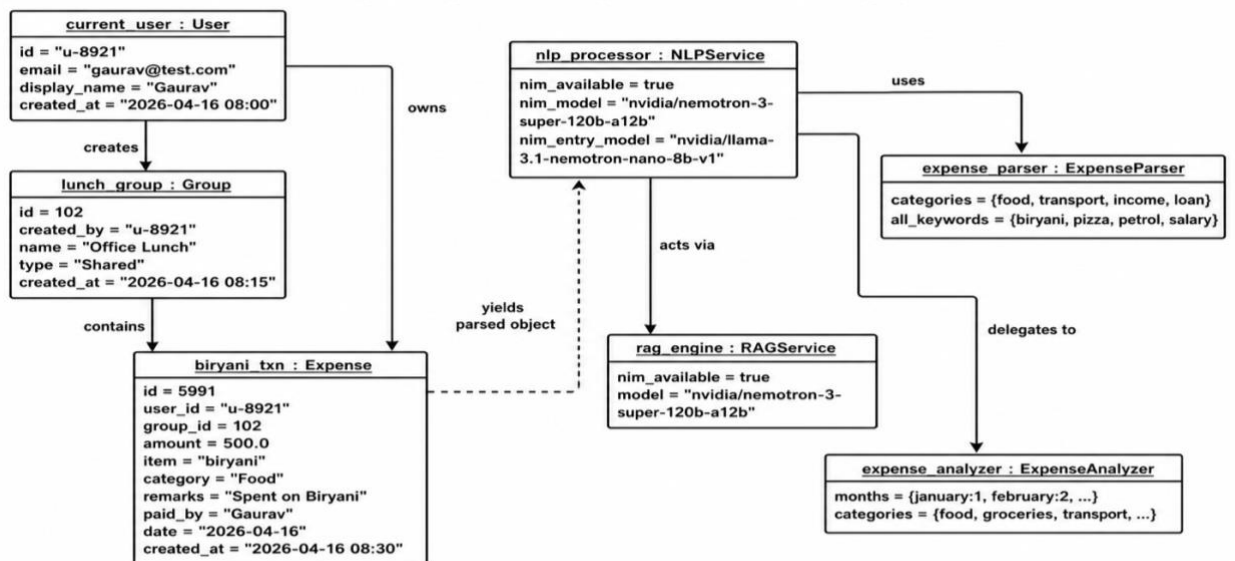


Figure 3. 5: Object Diagram of "PFM"

3.1.5 Dynamic Modelling: State & Sequence diagram

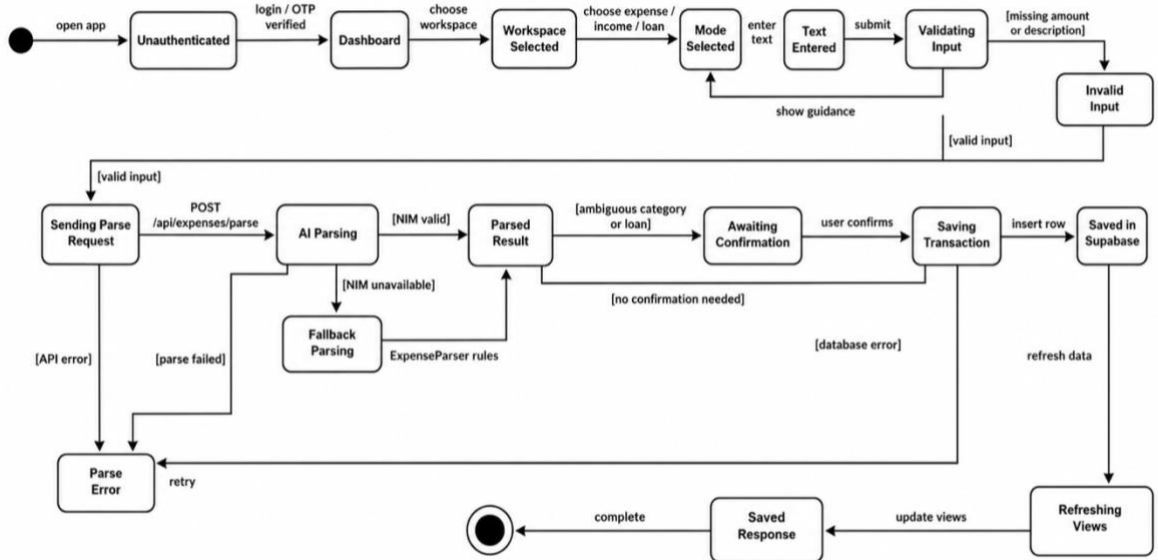


Figure 3. 6: State Diagram of "PFM"

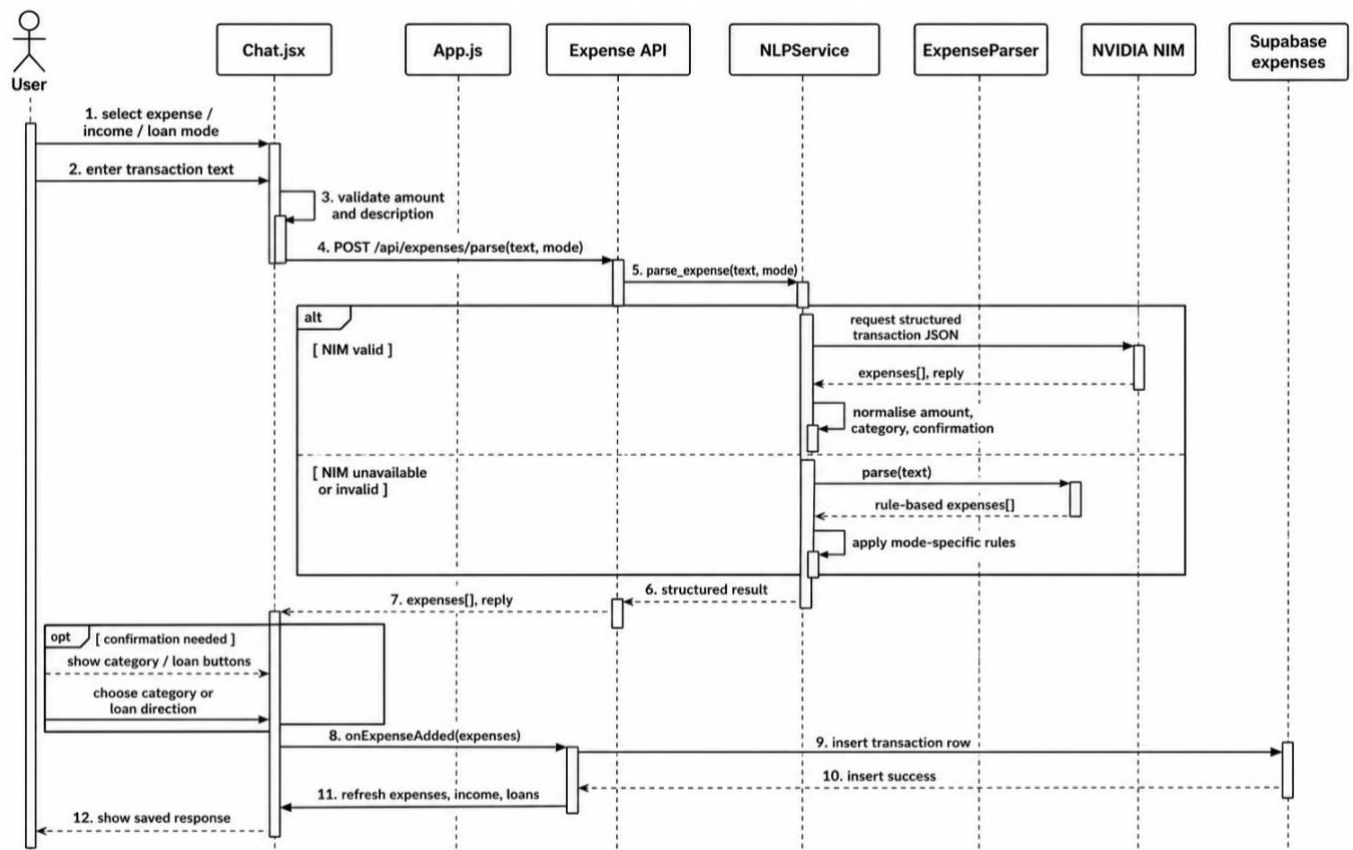


Figure 3. 7: Sequence Diagram of "PFM"

3.1.6 Process Modelling: Activity Diagram

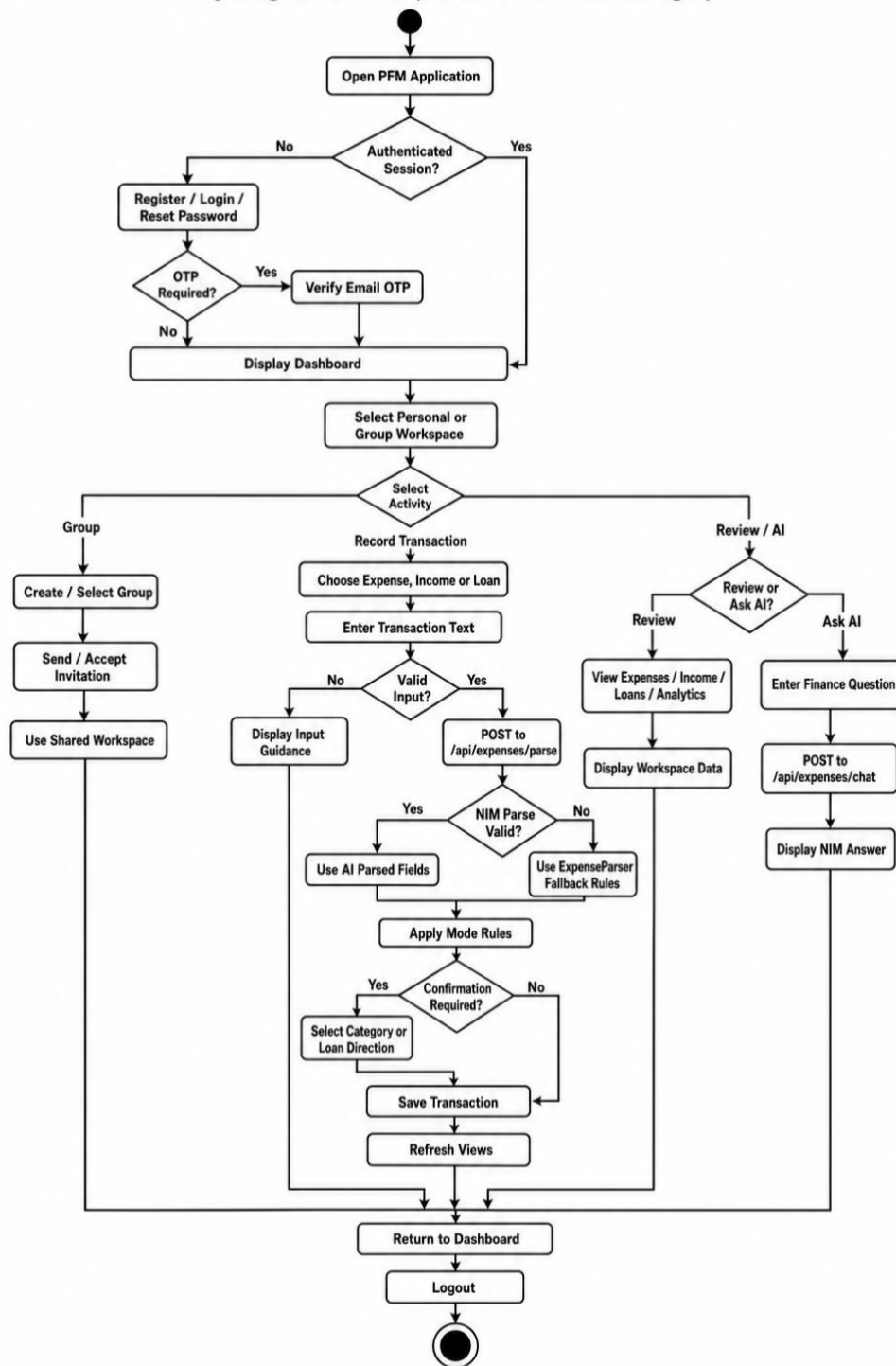


Figure 3. 8: Activity Diagram of "PFM"

3.2 System Design

3.2.1 Component Diagram

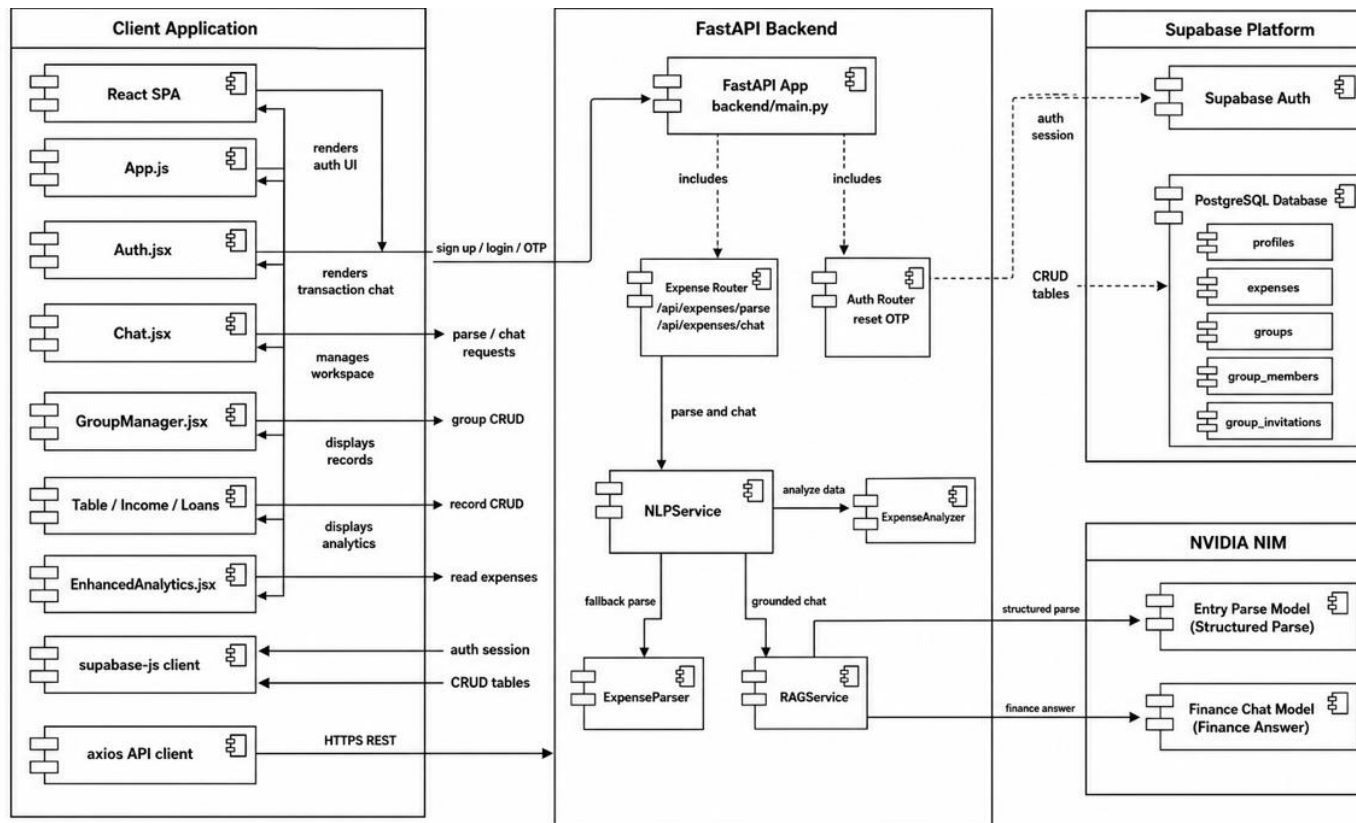


Figure 3. 9: Component Diagram of "PFM"

3.3 Algorithm Details

Algorithm: NLP Expense Parsing Algorithm

Function ParseExpense(inputText)

```
// Normalize the input text
```

```
text = toLowerCase(inputText)
```

```

// Extract amount using regular expression

amount = extractNumber(text)


// Define expense categories and keywords

categories = {

    "Food": ["lunch", "dinner", "food", "restaurant"],

    "Shopping": ["shopping", "clothes", "mall"],

    "Transport": ["bus", "taxi", "fuel", "ride"],

    "Utilities": ["electricity", "water", "internet"]

}


maxScore = 0

selectedCategory = "Others"


// Match keywords with categories

For each category in categories

    score = 0

    For each keyword in categories[category]

        If keyword exists in text

            score = score + 1

        End If

    End For

// Select category with highest score

```

```
    If score > maxScore
        maxScore = score
        selectedCategory = category
    End If
End For

// Return extracted details
Return(amount, selectedCategory)

End Function
```

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

The implementation phase involves transforming the proposed Personal Finance Manager (PFM) system into a functional web application. This phase includes developing the frontend interface, backend services, database integration, NLP-based expense parsing, budget optimization features, and conversational financial analysis modules. The implementation process ensures that the system fulfills the defined requirements and provides users with a smooth and efficient financial management experience.

4.1.1 Tools Used

- i. **Front End:** The frontend of the Personal Finance Manager is developed using React.js to create a responsive and interactive web interface. Tailwind CSS is used for styling to provide a clean and user-friendly design, while JavaScript manages interactivity and application logic. Recharts is integrated to visualize financial analytics and spending trends through charts and graphs.
- ii. **Back End:** The backend is developed using FastAPI, a modern Python framework used for building high-performance APIs and handling server-side operations. FastAPI manages request routing, authentication, NLP processing, and communication between the frontend and database. NIM API is integrated for AI-assisted expense categorization and conversational financial analysis.
- iii. **Database and Hosting:** Supabase PostgreSQL is used as the primary database for securely storing user information, transaction records, and financial data. Supabase authentication services are used for secure user login and data protection. The application can be deployed on cloud hosting platforms such as Vercel, Render, or Railway to ensure scalability and availability.
- iv. **CASE Tools:** Visual Studio Code (VS Code) is used as the primary Integrated Development Environment (IDE) for coding and debugging. Draw.io and Lucidchart are used for creating UML diagrams, flowcharts, ER diagrams, and system architecture designs, while Figma is used for designing user interfaces and application layouts.

4.1.2 Implementation Details of Modules

Different modules are integrated into the Personal Finance Manager (PFM) web application to handle key functionalities essential for smart financial management:

- **Registration Module:** This module allows new users to register by providing details such as name, email, and password. It validates user information and securely stores account data in the database.
- **Login Module:** Users can securely log into the system using their registered email and password. The system verifies credentials and provides access to personalized financial dashboards and features.
- **Expense Management Module:** This module enables users to add, edit, delete, and view expenses. Users can record expenses using natural language input, making financial tracking easier and more efficient.
- **NLP Expense Parsing Module:** This module processes user-entered expense descriptions using Natural Language Processing techniques. It extracts important details such as transaction amount, category, and description automatically using regular expressions and probabilistic keyword matching.
- **Budget Optimization Module:** The budget optimization module analyzes historical spending patterns and provides intelligent suggestions to help users reduce unnecessary expenses and achieve savings goals.
- **Dashboard and Analytics Module:** This module displays spending summaries, category-wise analytics, charts, and financial trends using graphical visualizations for better financial understanding.
- **Conversational Financial Assistant Module (RAG):** This module allows users to ask natural language questions regarding their expenses and financial history. Using Retrieval-Augmented Generation (RAG) and NIM API integration, the system retrieves relevant transaction data and generates context-aware responses.

4.2 Testing

The testing phase ensures that all modules of the “PFM” function correctly and meet the intended requirements.

4.2.1 Test Cases for Unit Testing

Table 1: Unit Test of ‘PFM’

S.N	Test Case	Test Input Data	Expected Result	Actual Outcome	Status
1	User Registration	Valid name, email, password	User account created successfully	Account created successfully	Pass
2	User Login	Valid email and password	User logged into dashboard	Login successful	Pass
3	Add Expense	“Paid 500 for lunch”	Expense added with correct category	Expense added as Food category	Pass
4	NLP Expense Parsing	“Bought groceries for 2000”	Extract amount and category automatically	Amount and category extracted correctly	Pass
5	Budget Optimization	Savings target input	Generate budget reduction suggestions	Suggestions generated successfully	Pass

6	Dashboard Analytics	User expense records	Display charts and summaries	Charts displayed correctly	Pass
7	Conversational Query	“How much did I spend on food?”	Return relevant financial response	Accurate response generated	Pass
8	Delete Expense	Selected expense record	Expense removed from database	Expense deleted successfully	Pass
9	Logout	User logout request	User session terminated	Logout successful	Pass

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION

5.1 Modules Completed

- **User Authentication:** Implemented secure user registration and login functionalities using Supabase authentication, allowing users to create accounts and access personalized financial data securely.
- **Expense Management Module:** Developed functionality for adding, editing, deleting, and viewing expenses through a simple and user-friendly interface.
- **NLP Expense Parsing Module:** Implemented Natural Language Processing (NLP) techniques to automatically extract expense details such as amount and category from natural language inputs.
- **Dashboard and Analytics Module:** Developed an interactive dashboard to display spending summaries, category-wise analytics, and financial trends using charts and visualizations.
- **Budget Optimization Module:** Implemented a greedy budget optimization algorithm that analyzes historical spending data and provides budget-saving suggestions.
- **Frontend Development:** Designed and developed a responsive user interface using React and Tailwind CSS to provide smooth user interaction across different devices.
- **Backend API Development:** Developed backend services using FastAPI to manage authentication, expense processing, database communication, and API routing.
- **Database Integration:** Integrated Supabase PostgreSQL database for secure storage and management of user data, expense records, and financial analytics.

5.2 Modules Remaining

- **Conversational Financial Assistant (RAG):** Complete the Retrieval-Augmented Generation (RAG) module to allow users to ask financial questions and receive context-aware responses based on stored transaction records.
- **Gemini API Optimization:** Improve AI-assisted expense categorization and response accuracy for ambiguous user inputs.
- **System Testing and Debugging:** Perform complete unit testing, system testing, and bug fixing to ensure system reliability and performance.

- Deployment and Performance Optimization: Optimize application performance and deploy the system to a cloud hosting platform for production use.
- Report Documentation and Final Presentation: Complete project documentation, prepare the final report, and create presentation materials for the final defense.

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